



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SAKTHI AUTO FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

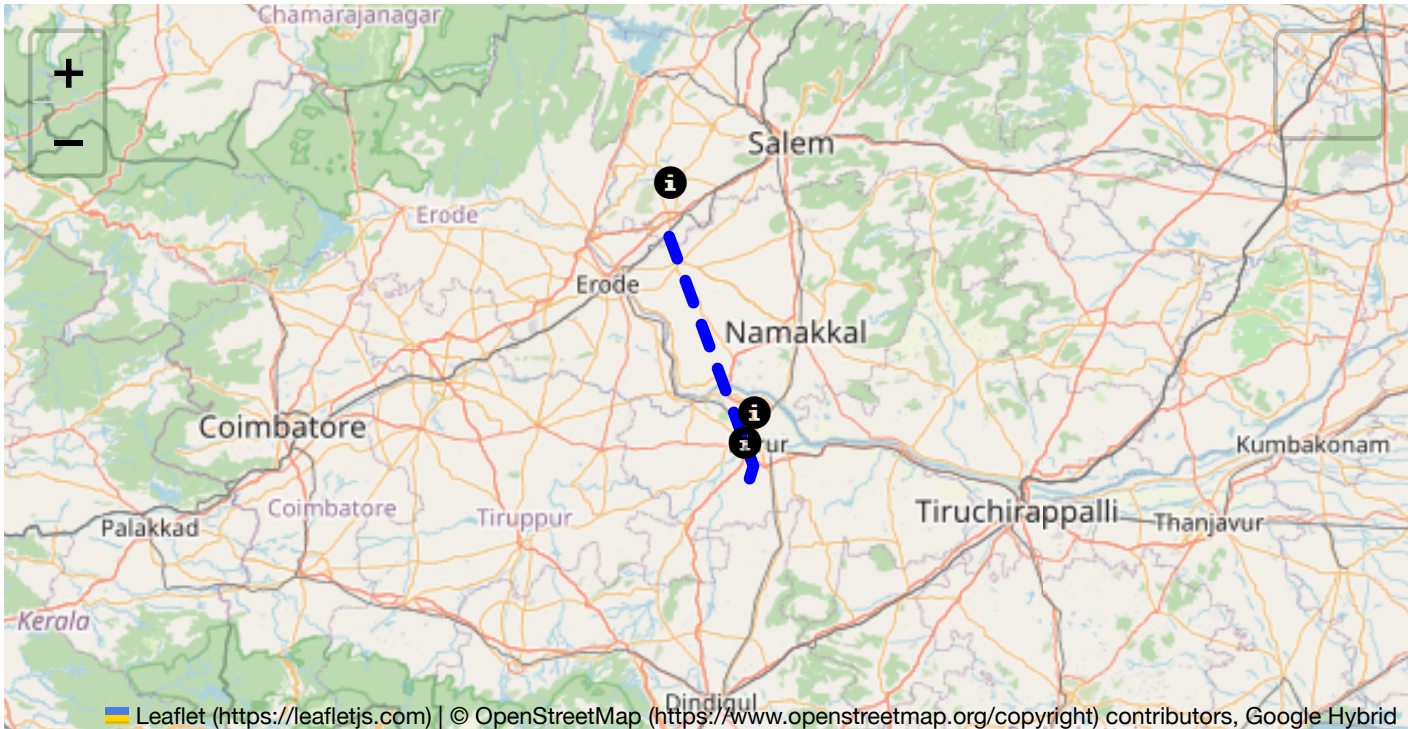
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

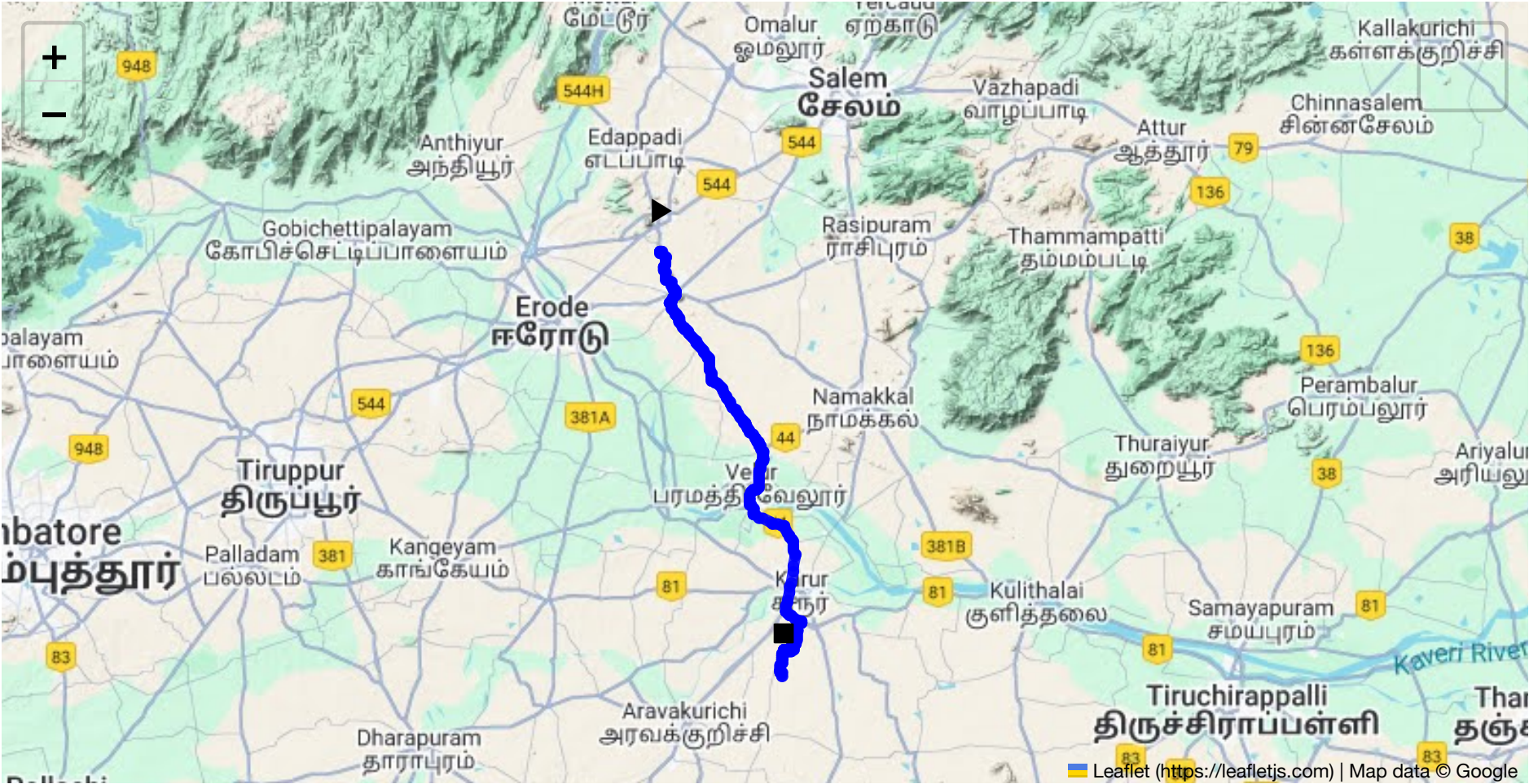
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 82.65 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours

Start: (11.4381, 77.8734)
End: (10.8377367, 78.048369)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Paganatham via Kodangipatti covers approximately 82.65 kilometers through Tamil Nadu. This journey typically takes around 1.66 hours for heavy vehicles, depending on traffic and road conditions.

Typical Weather Conditions

- **Climate:** Predominantly tropical with moderate to heavy rainfall during the monsoon (June to September).
- **Hazards:** Potential for reduced visibility and slippery roads during rains; heat waves may lead to road fatigue.
- **Advisory:** Monitor weather forecasts actively, especially during monsoon seasons to plan appropriately.

Traffic Patterns

- **Peak Congestion:** Morning (7 AM - 10 AM) and evening (5 PM - 8 PM) on weekdays.
- **Hotspots:** Urban areas and major intersections, especially near town centers like Sangari and Kodangipatti, tend to experience higher congestion.

Road Quality and Infrastructure

- **Condition:** Generally paved but varying quality; some stretches might be narrow without proper shoulder spaces.
- **Hazards:** Occasional potholes and less illuminated segments; rural stretches may lack clear signage.
- **Advisory:** Exercise caution, particularly in less developed stretches.

Alternative Routes for Emergencies

- **Detours:** When encountering major issues, consider secondary state roads. Always consult recent navigation updates as local redevelopment affects available alternatives.

Local Regulations on Hazardous Material Transport

- **Regulations:** Follow guidelines issued by the Tamil Nadu Pollution Control Board and Ministry of Road Transport for transport of hazardous materials, which include licensing, route permits, and safety signage requirements.
- **Checkpoints:** Be prepared for inspections at select check posts.

Historical Incidents

- **Data:** Limited public records on heavy vehicle accidents specific to hazardous materials. General road accident occurrences are more frequent in the less-developed road stretches.
- **Precautionary Measures:** Continuous vigilance while driving, especially on winding or rural roads.

Environmental Considerations

- **Sensitive Areas:** Some eco-sensitive stretches, particularly near water bodies or agricultural lands, require careful navigation to prevent spillage or disturbances.
- **Advisory:** Maintain safe speeds and avoid unnecessary heavy braking in these areas.

Communication Coverage

- **Network Strength:** Generally strong in urban centers. Possible dead zones in rural stretches, especially in less populated or topographically challenging areas.
- **Recommendations:** Equip vehicles with offline navigation devices and emergency communication tools.

Emergency Response Times

- **Urban Areas:** Faster response within 30-45 minutes.
- **Rural/Mid-Route Areas:** Could exceed 60 minutes due to less proximity to emergency services.
- **Advisory:** Establish quick contact points with local services along the route.

Risk Assessment Summary

- **Overall Risk:** Moderate, with specific considerations for weather-related challenges, road quality, and communication gaps.
- **Recommendations:** Ensure thorough pre-trip preparations including vehicle checks, communication setups, regulatory compliance, and maintaining an updated list of emergency contacts.

By adhering to these guidelines and maintaining awareness of identified risks, drivers can enhance their safety and ensure smoother transportation of hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Turn	High	10.91709, 78.07745	15 KM/Hr
13	Turn	Medium	10.91432, 78.07742	30 KM/Hr
14	Turn	Medium	10.91048, 78.07030	30 KM/Hr
15	Turn	High	10.89123, 78.07065	15 KM/Hr
16	Turn	High	10.89095, 78.06934	15 KM/Hr
17	Turn	Medium	10.88579, 78.06845	30 KM/Hr
18	Turn	High	10.88516, 78.06803	15 KM/Hr
19	Turn	Medium	10.88473, 78.06595	30 KM/Hr
20	Turn	High	10.88482, 78.06494	15 KM/Hr
21	Turn	Medium	10.87764, 78.06542	30 KM/Hr
22	Turn	High	10.87589, 78.06542	15 KM/Hr
23	Turn	High	10.87606, 78.05213	15 KM/Hr
24	Turn	Medium	10.86360, 78.04922	30 KM/Hr
25	Turn	Medium	10.86328, 78.04879	30 KM/Hr
26	Turn	High	10.84759, 78.04652	15 KM/Hr
27	Turn	High	10.84731, 78.04767	15 KM/Hr
28	Turn	High	10.84353, 78.04787	15 KM/Hr
29	Turn	High	10.84318, 78.04886	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
19	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
21	hospital	Government Hospital, Kodangipatti	10.91425, 78.078034	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
20	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.92958, 78.05452



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.91709, 78.07745



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91432, 78.07742



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91048, 78.07030



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.89123, 78.07065



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.89095, 78.06934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88579, 78.06845



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.88516, 78.06803



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88473, 78.06595



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.88482, 78.06494



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.87764, 78.06542



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87589, 78.06542



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87606, 78.05213



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.86360, 78.04922



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.86328, 78.04879



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84759, 78.04652
